

(ii) State the **two** variables that need to be controlled in this experiment. [2]

1

2

(iii) State **one** reason why these variables need to be controlled. [1]

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(b) In a different experiment hydrochloric acid was poured into a flask containing different volumes of sodium thiosulfate solution and water. The time taken to react was measured. The results are shown in the table below.

Volume of acid (cm ³)	Volume of sodium thiosulfate (cm ³)	Volume of water (cm ³)	Time for cross to disappear from view (s)
60	55	5	44
60	50	10	51
60	45	15	57
60	40	20	83
60	35	25	106

(i) State what conclusion can be made from the results. [1]

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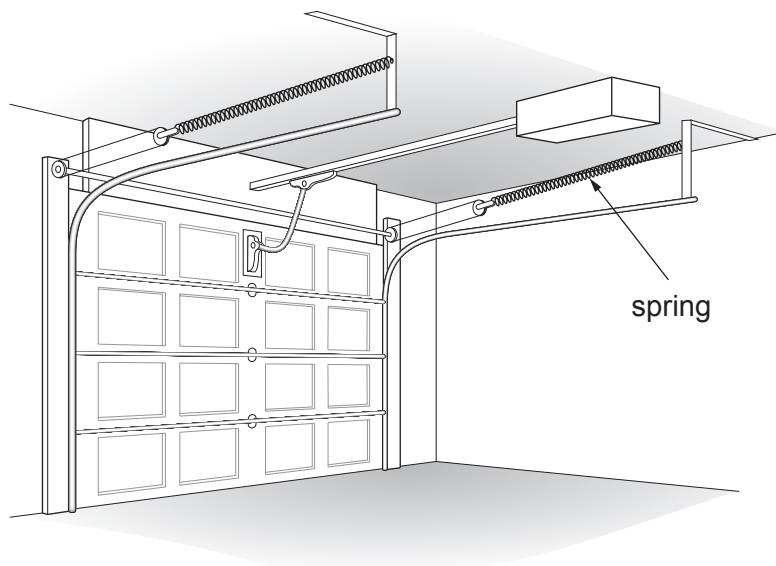
(ii) Explain the results in terms of particles. [2]

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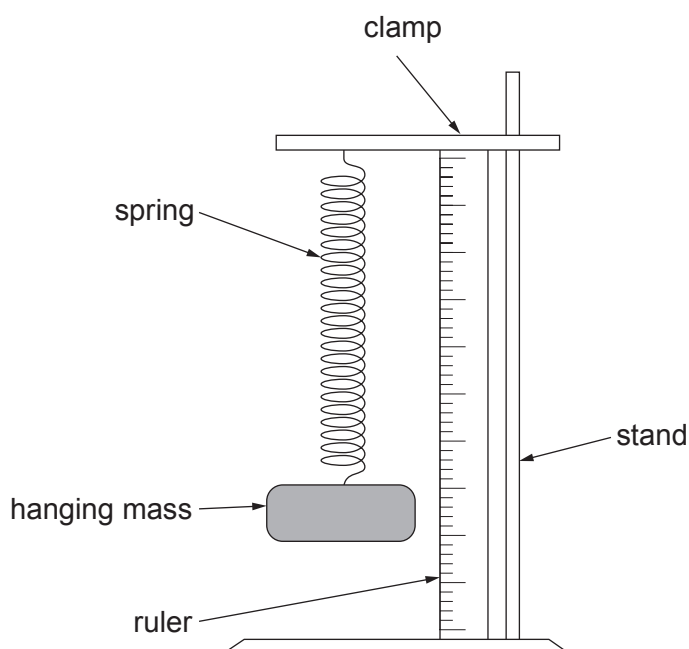
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6. Springs can be used in garage doors. When the door shuts the springs are stretched and put under tension. When the door opens the spring returns to its original length making it easier to raise.

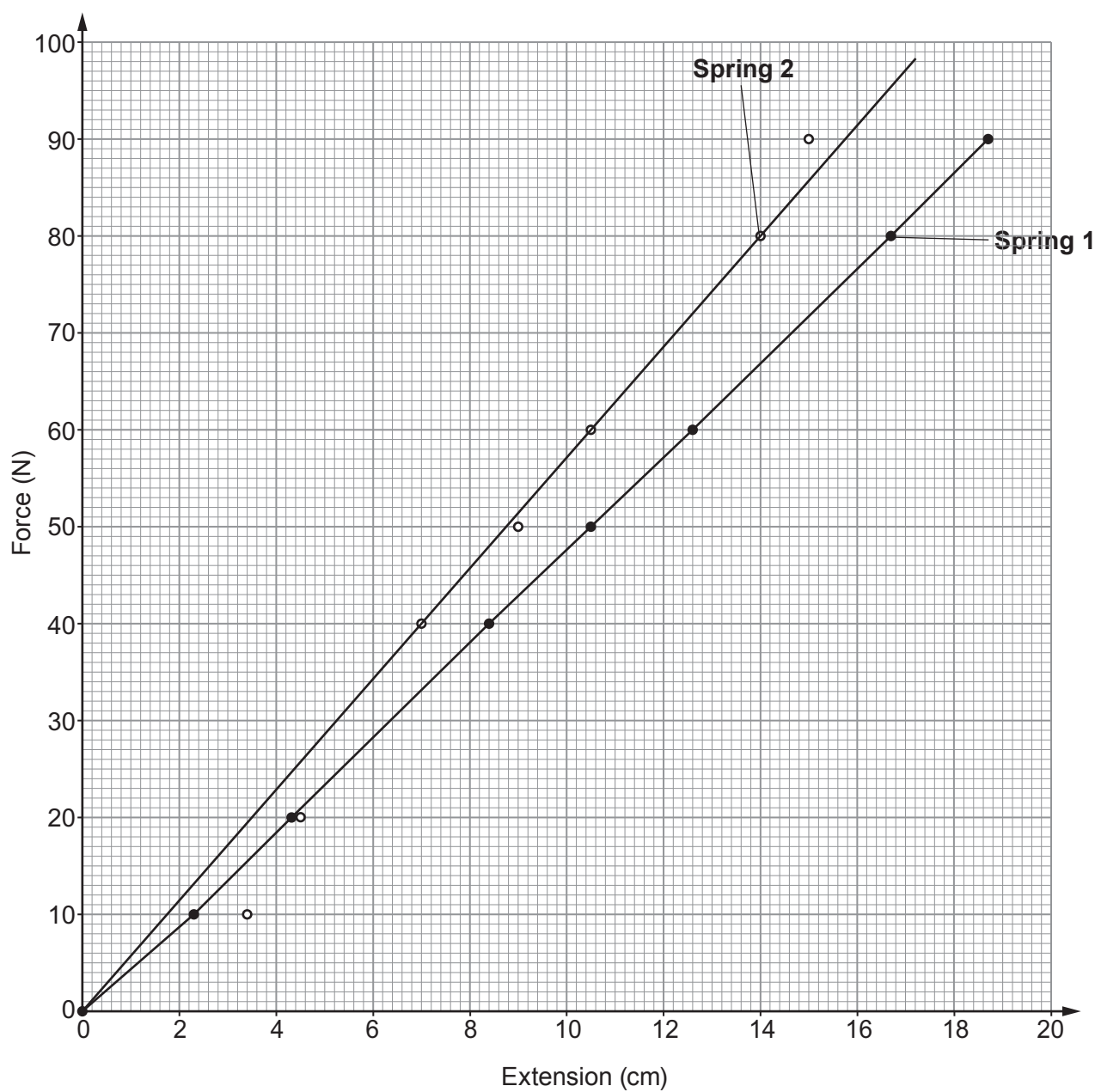


It is important that the spring is designed so that it does not become permanently stretched during the process. Springs become permanently stretched once they pass their **elastic limit**. This is the point where extension is no longer proportional to the stretching force.

Students test three springs using the apparatus shown below.



(a) Their results for two springs used in the doors are shown in the graph below.



- (i) Compare the behaviour of the springs as the force increases from 10 N to 40 N.

[2]

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- (ii) Use the graph and the equation:

$$\text{spring constant} = \frac{\text{force}}{\text{extension}}$$

to calculate the value of the spring constant for **Spring 1**.

[3]

spring constant = N/cm

- (iii) Use the graph to state **one** reason why the students can have more confidence in the results for **Spring 1** than for **Spring 2**.

[1]

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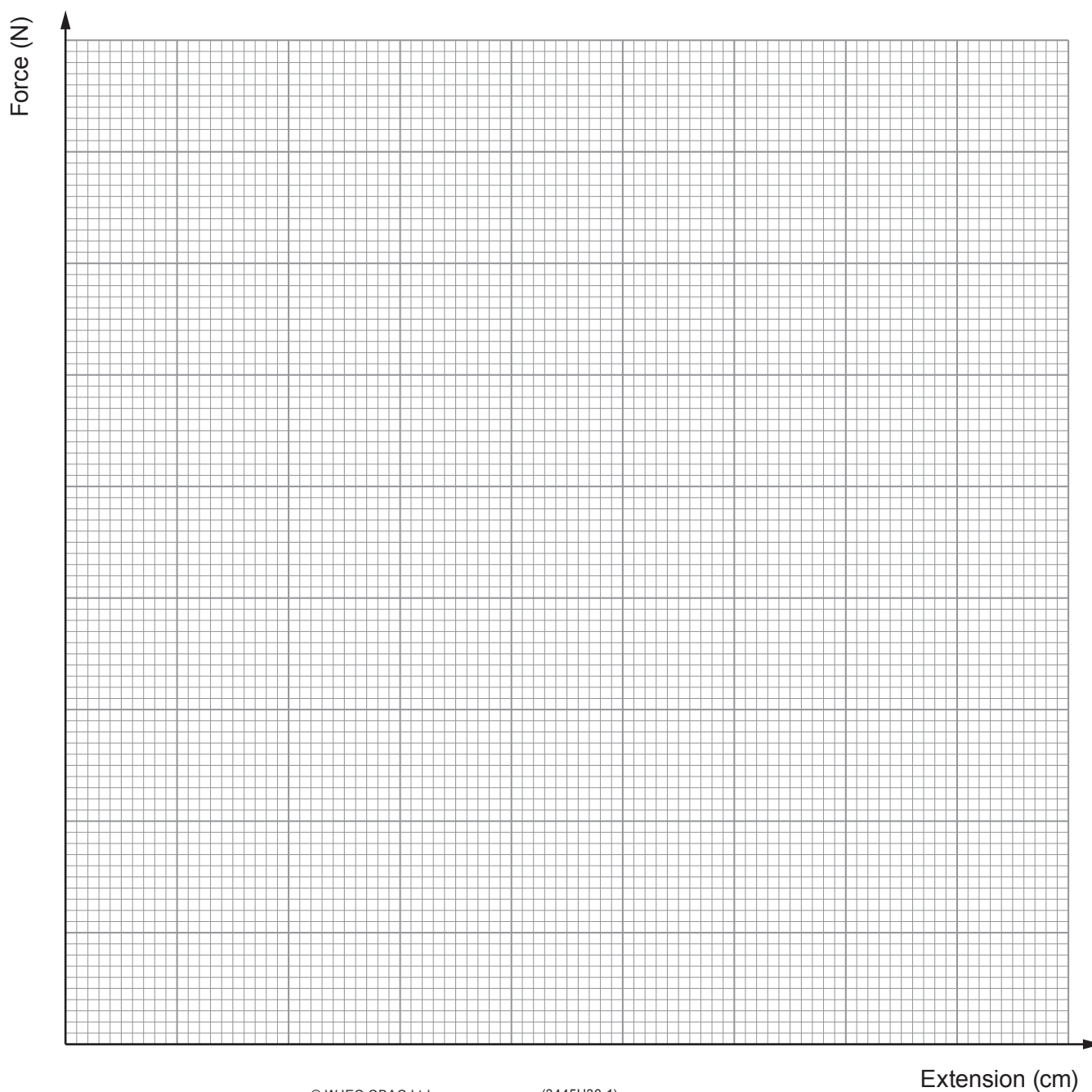
(b) The results for **Spring 3** are given in the table below.

Examiner
only

Force (N)	Extension (cm)
0	0.0
10	2.4
20	4.8
40	9.6
50	12.0
60	16.0
80	28.0
90	36.0

(i) Plot the data on the grid below and draw a suitable line.

[4]



- (ii) Springs become permanently stretched once they pass their **elastic limit**. This is the point where extension is no longer proportional to the stretching force.

I. **Label** the elastic limit on your graph with a letter **E**. [1]

II. Use your graph to find the force required to reach the elastic limit. [1]

Force = N

- (iii) During the opening and closing of the garage doors the spring will extend by 30 cm. Explain whether or not **Spring 3** is suitable for this purpose. [2]

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Answer **all** questions in the spaces provided.

1. Drinks manufacturers use standard food colourings which are identified by E-numbers. A student was asked to investigate which E-numbers were present in different brands of orange drinks.

The student suggested the following method:

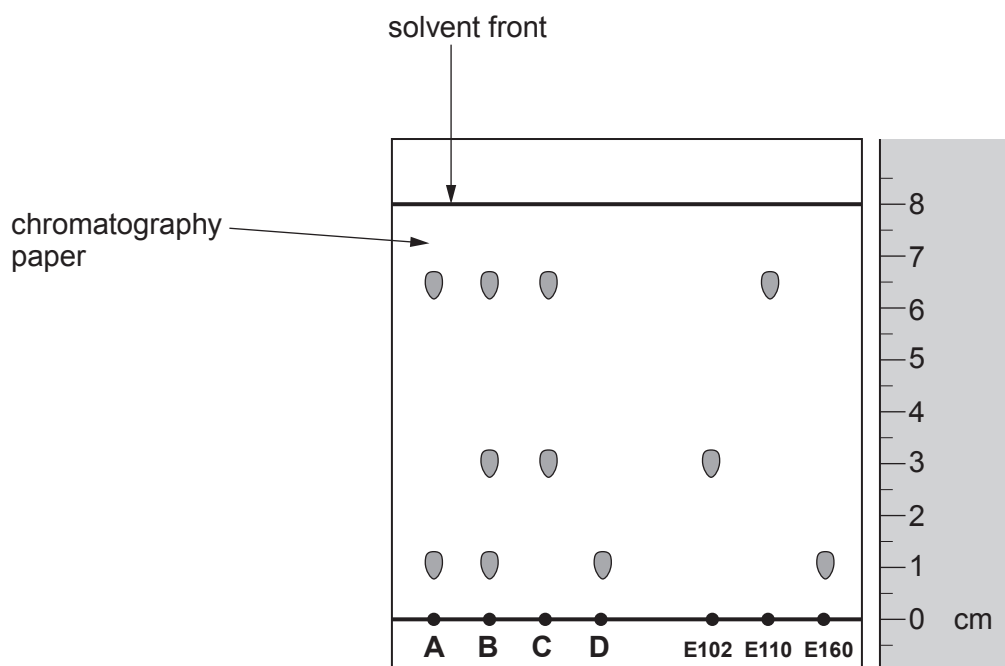
1. Place spots of each drink **A**, **B**, **C** and **D** on a pen line on chromatography paper.
2. Place spots of three common food colourings, **E102**, **E110**, **E160** on the pen line.
3. Hang the chromatography paper in a beaker and add water until it is above the pen line.
4. Leave the paper to stand in water in a beaker until the water rises to near the top of the chromatography paper.

- (a) The student failed to produce a chromatogram using this method. State **two** errors in this method. [2]

1.

2.

- (b) Another student, using a valid method, obtained the chromatogram below (*not drawn to scale*).



Answer the following questions using the information in the chromatogram.

- (i) State which food colouring (**E102**, **E110**, **E160**) is the most soluble. [1]

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- (ii) State which drinks (**A**, **B**, **C** or **D**) contained **E110**. [1]

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- (iii) The student predicted that all orange drinks contain the same E-numbers. Explain whether this prediction was correct by comparing the food colourings found in drinks **B** and **D**. [2]

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- (iv) The retention factor (R_f) value of a substance can be used to identify that substance from a chromatogram.

- I. State the distance travelled by the solvent front. [1]

Distance = cm

- II. Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by the solvent front}}$$

- to calculate the R_f value for **E102**. [2]

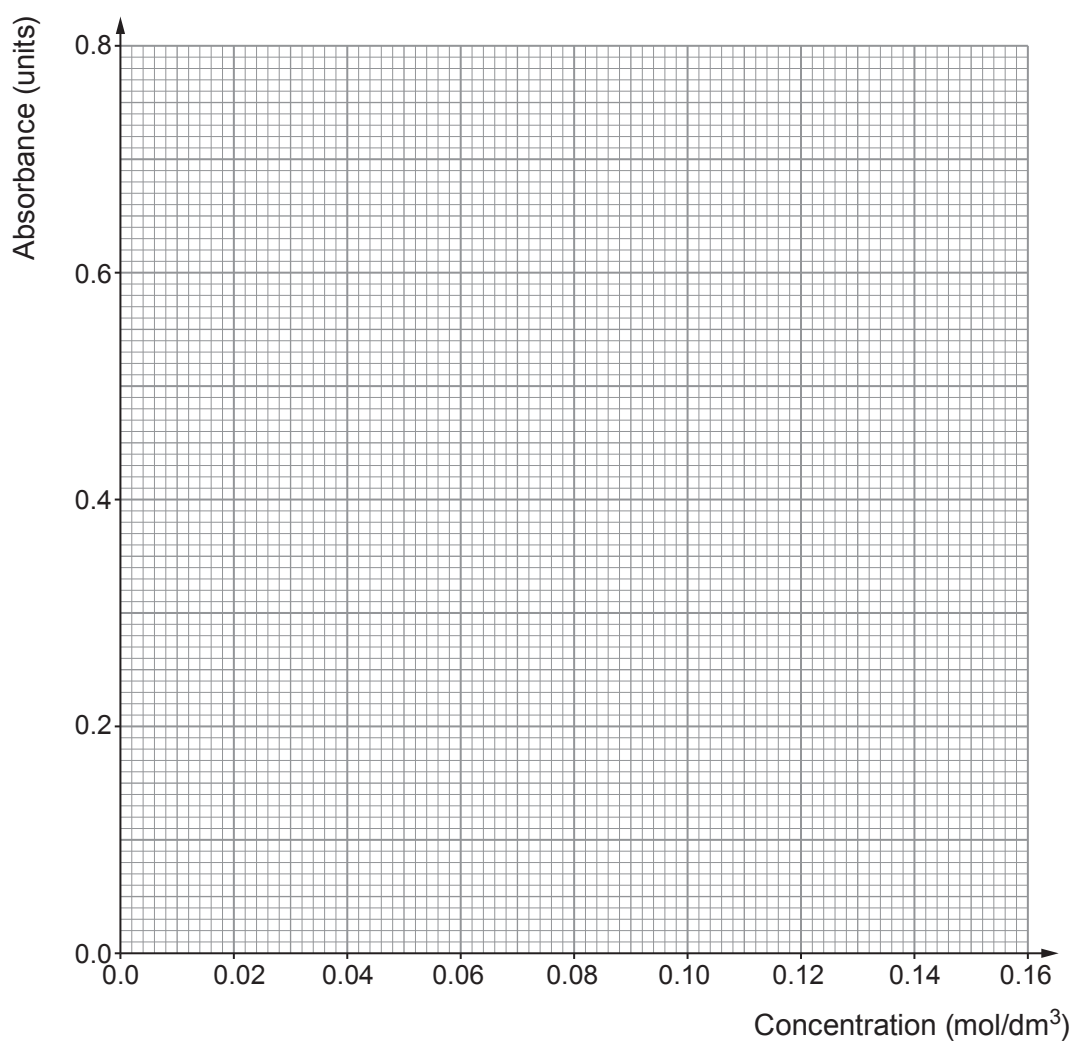
R_f =

3. An experiment was carried out to find the concentration of beta-carotene in a solution using colorimetry. The results are shown in the table below.

Concentration (mol/dm ³)	Absorbance (units)
0.02	0.1
0.04	0.2
0.08	0.4
0.12	0.6
0.16	0.8

- (a) Use the data to plot a graph on the grid below and draw a suitable line.

[3]



- (b) Describe how the absorbance of beta-carotene changes with concentration.

[1]

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- (c) Use your line to predict the concentration of beta-carotene with an absorbance of 0.3 units.

[1]

Concentration = mol/dm³

- (d) It is claimed that if the concentration is doubled then the absorbance always doubles. Explain whether you agree with this statement using data from the table.

[2]

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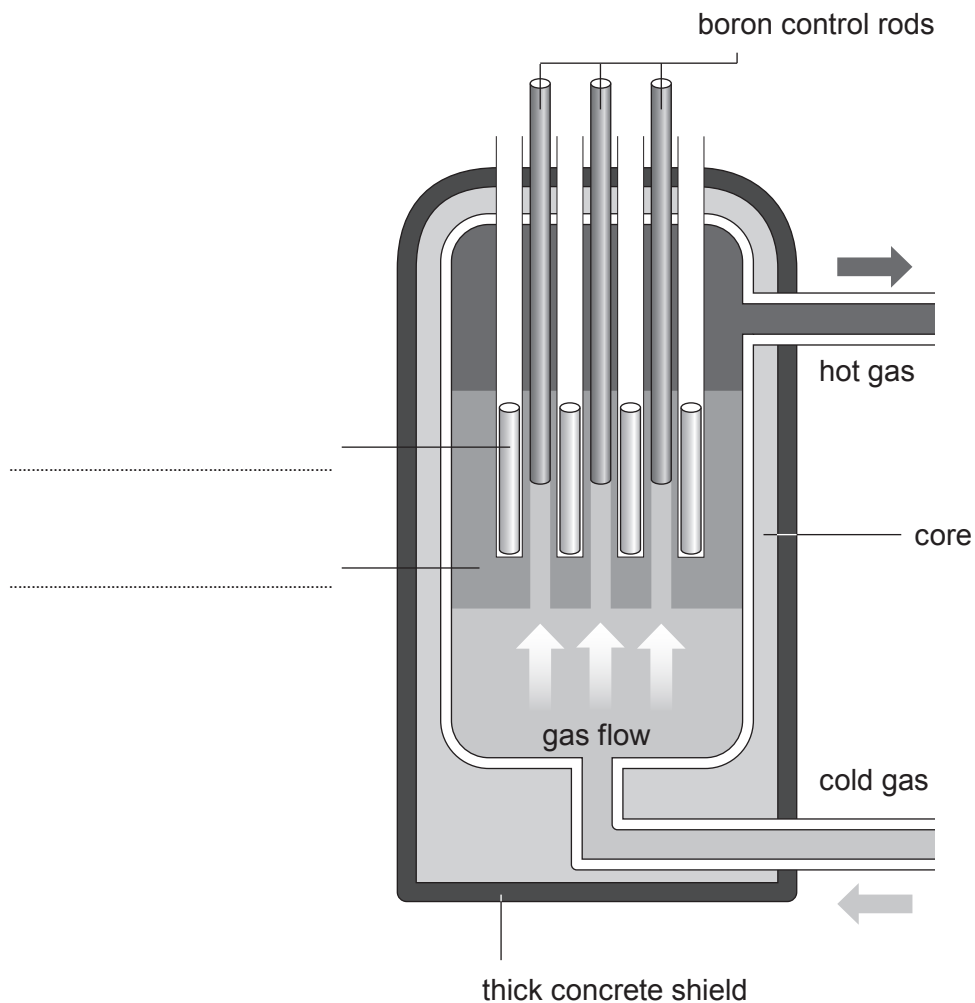
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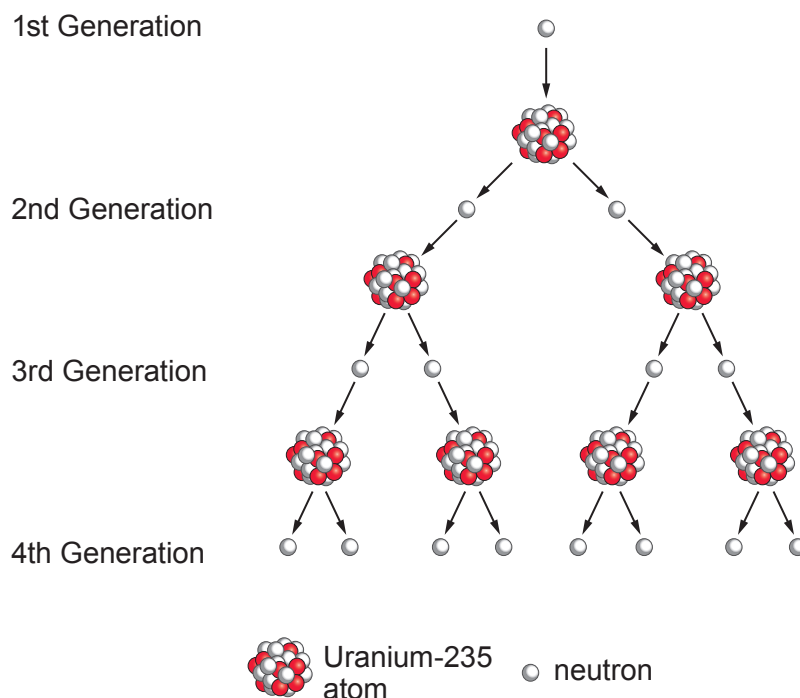
4. Nuclear fission occurs when a nucleus of uranium is split after absorbing a neutron. It is the process used in a nuclear reactor like the one shown in the diagram below.



(a) Complete the labelling of the diagram.

[2]

- (b) The diagram below shows an uncontrolled chain reaction of the decay of uranium.



- (i) Calculate the number of 5th generation neutrons. [1]
- (ii) Explain how the number of neutrons in your answer above would change if the control rods are lowered further into the reactor. [2]

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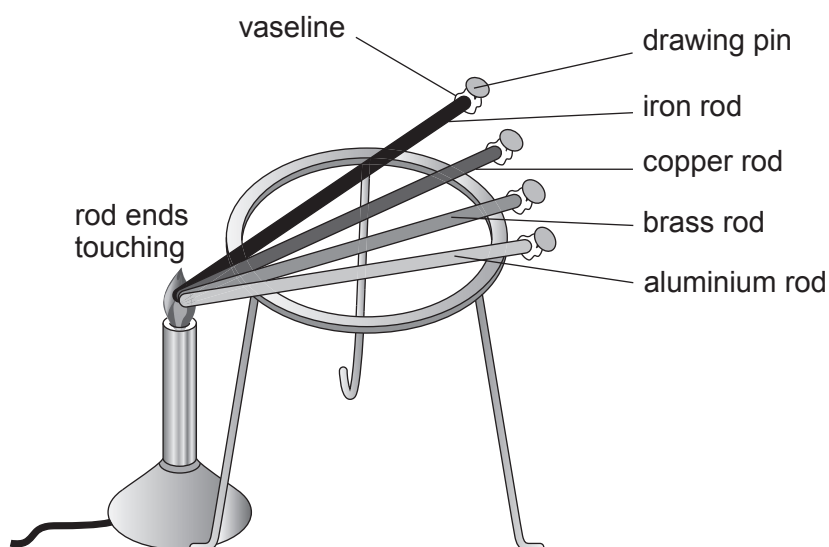
- (iii) State the danger of an uncontrolled chain reaction in a nuclear reactor. [1]

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- [6 QÉR]

2. Aluminium, copper and iron are metals. Brass is a metal alloy. These materials have many uses.

The thermal conductivity of metals can be investigated using the apparatus below. The time taken for a drawing pin to fall from each metal rod is measured.



- (a) (i) State the independent variable in this experiment. [1]

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- (ii) State **two** variables that are controlled in this experiment. [2]

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- (iii) Describe how the thermal conductivity is related to the dependent variable. [2]

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- (b) Two students obtained the results shown in the table below.

Metal	Time taken for pin to drop (s)
iron	17
copper	7
brass	12
aluminium	5

- (i) Use the results shown in the table above to place iron, copper, brass and aluminium in order of conductivity. [1]

Best conductor



Poorest conductor

- (ii) The following table shows the theoretical values of the thermal conductivity of the metals.

Metal	Thermal conductivity (units)
iron	80.4
copper	401
brass	109
aluminium	210

Use the information above to explain whether the theoretical data agrees with experimental data. [3]

Best conductor



Poorest conductor

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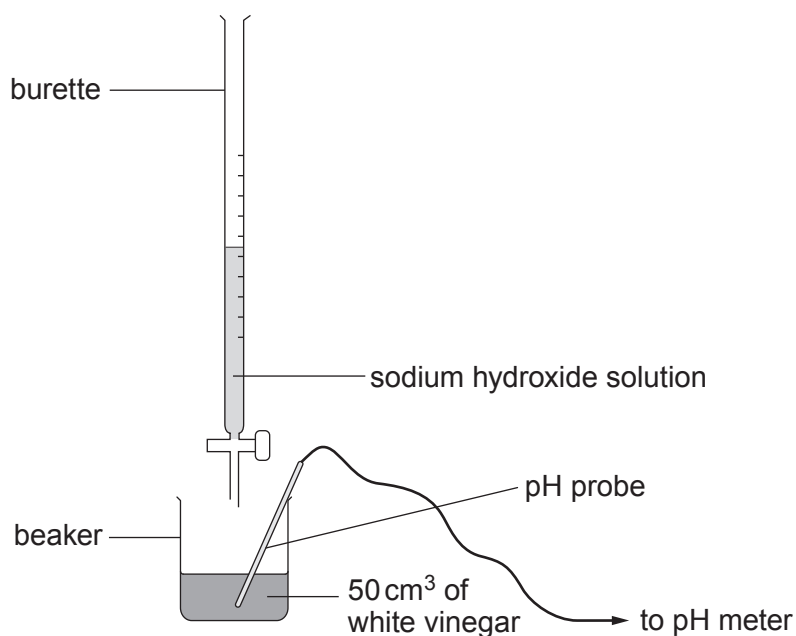
- (c) Mercury is a metal which has a melting point of -39°C . Suggest why the students could not use mercury in this investigation. [1]

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- (b) It is suspected that supplies of vinegar have been tampered with by adding a strong acid. White vinegar usually contains ethanoic acid at a concentration of 0.01 mol/dm^3 . Samples of the vinegar are taken and tested by titrating against 0.5 mol/dm^3 sodium hydroxide solution using the apparatus below.



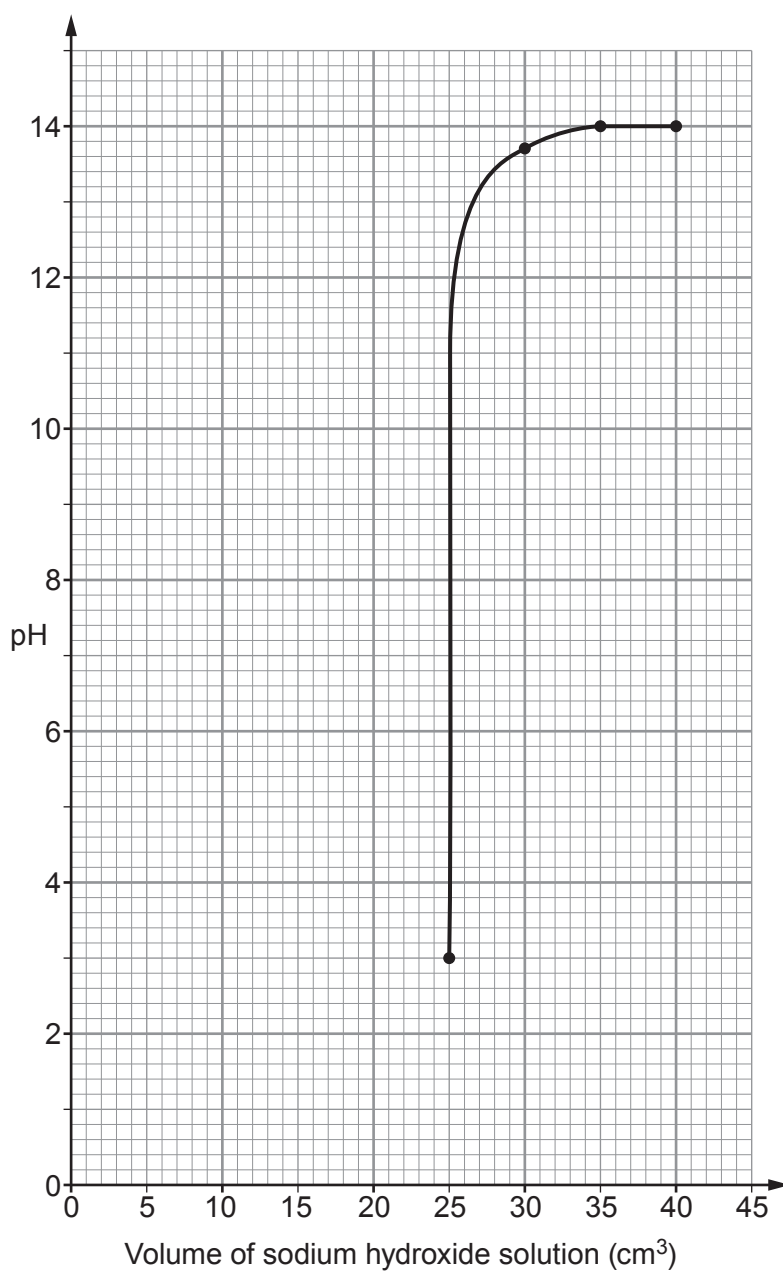
A pH meter is used to monitor the pH of the solution in the flask. The results are shown in the table below.

Volume of sodium hydroxide solution (cm^3)	pH reading
0	1.5
5	1.6
10	1.7
15	1.9
20	2.2
25	3.0
30	13.7
35	14.0
40	14.0



- (i) Use the data in the table to complete the graph on the grid below.

[3]



- (ii) Use the graph to find the volume of sodium hydroxide solution required to neutralise the white vinegar.

[1]

volume = cm³

- (iii) Use the information on pages 16 and 17 and the equation

$$\text{concentration of white vinegar} = \frac{\text{concentration of sodium hydroxide} \times \text{volume of sodium hydroxide}}{\text{volume of white vinegar}}$$

to determine whether or not the samples had been tampered with. [3]

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- (iv) Complete the word equation for the reaction between an acid and a base. [2]

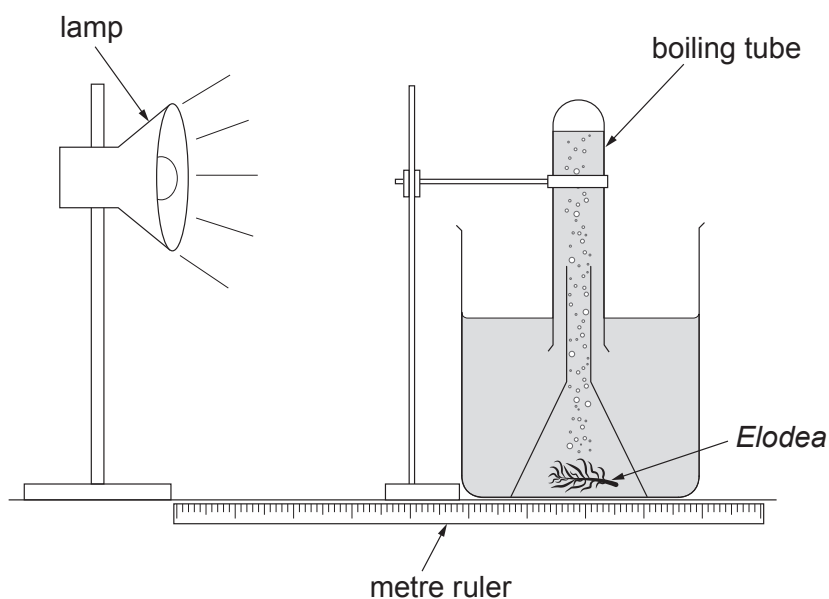
acid + base →

- (v) Explain why taking multiple readings will increase the accuracy of the result. [2]

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- (b) The following apparatus is used to investigate how light intensity affects the rate of photosynthesis of an *Elodea* plant.



The method used is described below.

1. Add **one spatula** of sodium hydrogen carbonate to a beaker containing 200 cm^3 of water at 20°C , to supply carbon dioxide.
2. Place the *Elodea* in the beaker.
3. Completely fill a boiling tube with water and carefully place over the end of the funnel with the end under the water, clamp into place.
4. Place the lamp 5 cm away from the *Elodea*.
5. Start the stopwatch and record the **number of bubbles** of oxygen produced in a **few seconds**.
6. Repeat the experiment with the lamp 10 cm, 15 cm, 20 cm, 25 cm and 30 cm from the apparatus.

- (i) Steps 1 and 5 could lead to uncertainties. In the table below, circle **one** improvement in each row that you would use.

[3]

Current method	Improvement	
add one spatula	add 1 kg	add 1 g
record the number of bubbles	collect the gas in a beaker	collect the gas in a measuring cylinder
produced in a few seconds	produced in 1 minute	produced in 1 hour

- (ii) State how you would change step 6 to investigate the effect of temperature on photosynthesis.

[1]

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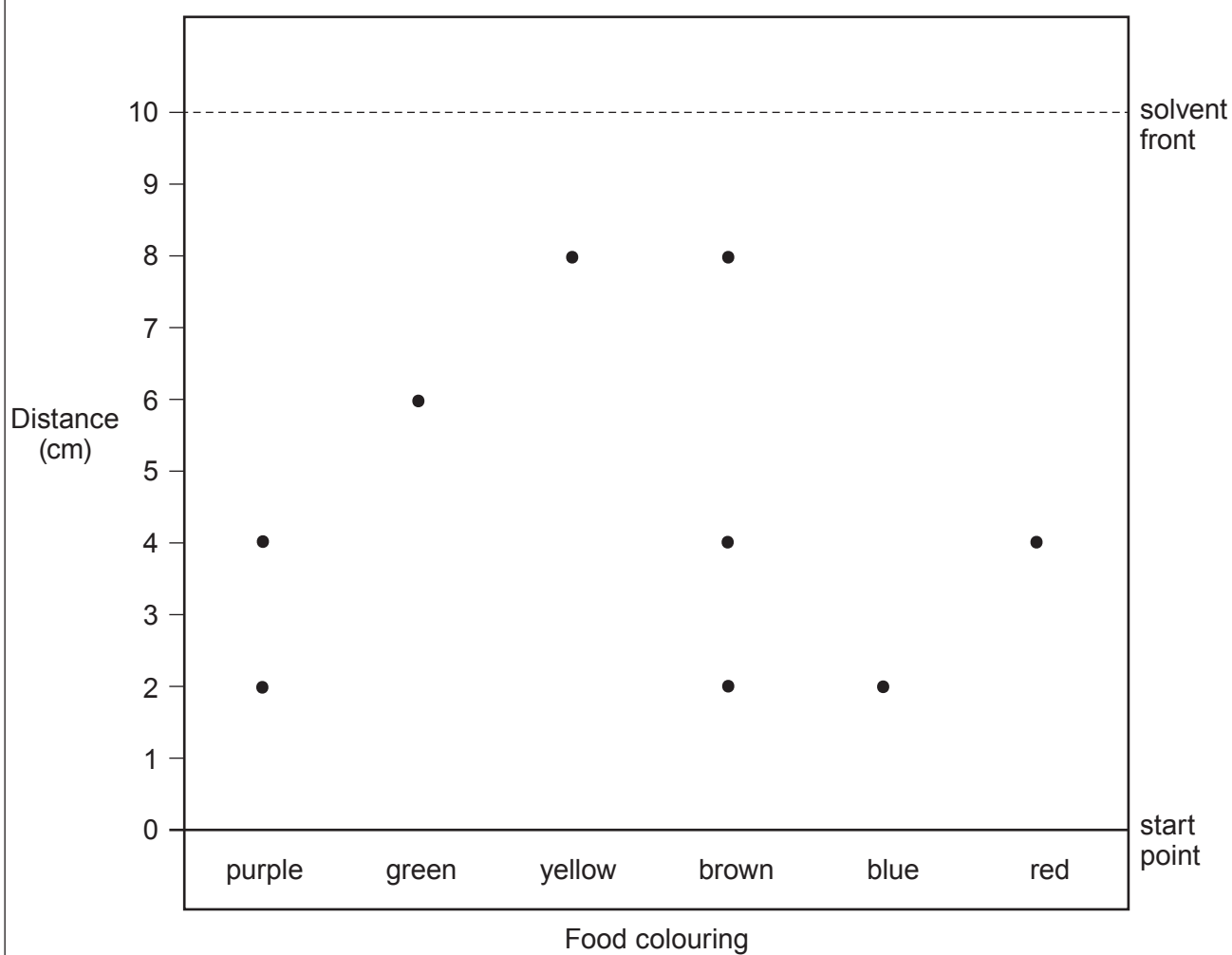
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2. Analytical scientists work in a wide range of different industries and agencies.

- (a) The Food Standards Agency (FSA) funded a project to analyse the colours used in food and drinks.

The diagram below shows a chromatogram of the dyes present in different food colourings.



- (i) It was thought that the dye in the blue colouring would not be present in other colourings.
Explain whether you agree.

[2]

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(ii) Use the equation:

$$R_f = \frac{\text{distance travelled by substance}}{\text{distance travelled by solvent}}$$

to calculate the R_f value for the dye in the **red** colouring.

[3]

$R_f =$

- (b) Analytical scientists also identify unknown compounds.
There were three bottles, each containing a solution of one of the following compounds.

copper(II) sulfate

lithium carbonate

sodium chloride

- (i) State which of these compounds would produce a yellow flame in a flame test. [1]

- (ii) I. State which of these compounds would produce bubbles of gas when added to hydrochloric acid. [1]

- II. The gas turns limewater milky. Name this gas. [1]

- (iii) 1 cm³ of silver nitrate solution is added to three different test tubes, each containing a solution of one of the compounds in the box above.

- I. State which of these compounds will give a white precipitate. [1]

- II. Complete the risk assessment below for silver nitrate. [2]

Hazard	Risk	Control measure
silver nitrate is irritant	could cause eye damage if splashes in the eyes when	
		
		
		



4. Heating is a common method used to preserve food.

- (a) **State three** methods, **other than heating**, in which the growth of bacteria can be slowed down or stopped in food products.

Explain how each method works.

[6 QER]

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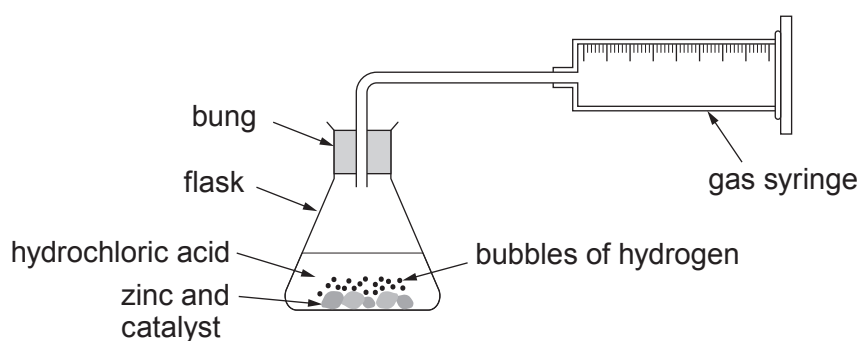
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5. The diagram below shows apparatus that can be used to investigate the rate of the reaction between large zinc pieces and hydrochloric acid. A catalyst was also added to the flask.

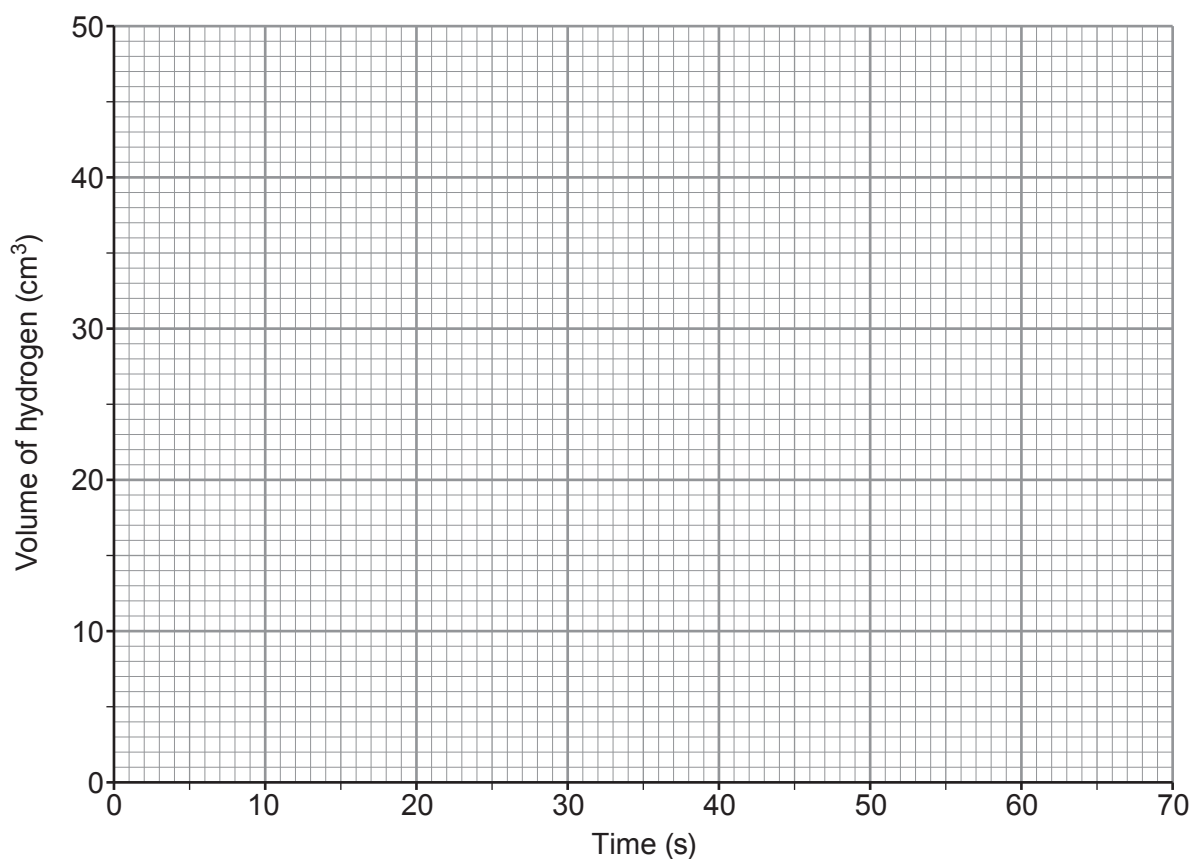


Large pieces of zinc were placed in excess dilute hydrochloric acid and the volume of hydrogen produced was recorded every 15 seconds. The results obtained are shown below.

Time (s)	0	15	30	45	60
Volume of hydrogen (cm ³)	0	28	40	48	48

- (a) (i) Plot the results on the grid below and draw a suitable curve.

[3]



(ii) Use your graph to answer the following questions.

I. Describe the relationship between volume of hydrogen produced and time.

[2]

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II. State the volume of hydrogen collected at 25 s.

[1]

volume = cm³

III. At what time did the reaction stop?

[1]

time = s

(b) (i) State the purpose of the catalyst.

[1]

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(ii) The mass of catalyst added at the start of the experiment was 1.5 g.
State the mass of catalyst remaining after 60 s.

[1]

mass = g

(iii) State how the **volumes in the table** would be different if the catalyst had not been added.

[1]

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(c) (i) State how the **volumes in the table** would be affected by using the same mass of zinc **powder** instead of large pieces of zinc.

[1]

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(ii) Explain how the **volumes in the table** would be affected if the original experiment was repeated at a higher temperature.

[2]

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